Radiomics and Deep Learning for Tumor Diagnosis and Prognosis: A Survey

Radiomics and deep learning aim to convert tumor imaging into reproducible biomarkers for diagnosis, grading, prognosis, recurrence prediction, treatment-response assessment, and therapy planning. Radiomics pipelines extract engineered intensity, texture, shape, and wavelet features, while deep models learn representations directly from CT, MRI, PET, ultrasound, mammography, histopathology, and multimodal clinical data. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around feature engineering, deep representation learning, diagnostic classification, survival and recurrence modeling, response prediction, radiogenomics and pathomics, imaging modalities and tumor sites, and clinical validation. The central argument is that clinically useful tumor imaging AI requires endpoint-aware modeling, careful validation, calibrated risk estimates, interpretable biomarkers, and integration with pathology, genomics, and treatment context.

Keywords:

radiomics; deep learning; tumor diagnosis; cancer prognosis; survival prediction; treatment response; radiogenomics; medical imaging

1 Introduction

Tumor imaging is central to oncology because it captures anatomy, phenotype, burden, heterogeneity, vascularity, and treatment response at scales that cannot be sampled exhaustively by biopsy. Radiomics and deep learning seek to make this imaging signal quantitative. Radiomics converts regions of interest into engineered descriptors such as intensity, shape, texture, wavelet, and spatial heterogeneity features. Deep learning learns representations from pixels, voxels, patches, volumes, or multimodal patient records. Both paradigms aim to support diagnosis, grading, prognosis, recurrence prediction, treatment-response assessment, and therapy planning [23, 46, 91, 130, 134, 184, 216, 237].

The field is clinically attractive because many tumor decisions are already image-driven. Radiologists assess lesion detectability, staging, progression, and response; oncologists combine imaging with histology, genomics, laboratory values, and therapy history; surgeons and radiation oncologists depend on segmentation and risk stratification. Imaging AI can therefore add value if it is reliable at the same decision points: differentiating benign from malignant lesions, classifying tumor subtype or grade, predicting survival and recurrence, forecasting response to chemotherapy, immunotherapy, radiotherapy, or neoadjuvant treatment, and identifying patients who need intensified surveillance [91, 106, 121, 134, 184, 211, 216, 237].

Yet the promise is difficult to realize. Radiomics features are sensitive to acquisition protocol, reconstruction, scanner vendor, segmentation, resampling, binning, and feature-selection choices. Deep models can learn powerful representations but may exploit site-specific shortcuts or fail under distribution shift. Prognostic endpoints are censored, imbalanced, and confounded by treatment. Response labels are delayed and sometimes inconsistent. Tumor biology is spatially heterogeneous, while many pipelines reduce imaging to a single region or score. These tensions make validation, calibration, interpretability, and multimodal integration as important as architecture design [26, 91, 105, 121, 130, 184, 211, 237].

This survey reviews radiomics and deep learning for tumor diagnosis and prognosis. We treat radiomics, deep learning, segmentation, radiogenomics, pathomics, multimodal fusion, survival modeling, and response prediction as parts of one translation pipeline. The goal is not only to catalog models, but to clarify where each modeling choice enters the clinical evidence chain.

2 Background

2.1 Radiomics as Quantitative Tumor Phenotyping

Radiomics pipelines usually begin with image acquisition and tumor or organ-at-risk segmentation. Images are normalized, resampled, discretized, and transformed; features are then extracted from first-order intensity statistics, shape, texture matrices, filters, and wavelet decompositions. Feature selection, harmonization, and machine-learning models convert high-dimensional descriptors into diagnostic labels, risk groups, or time-to-event predictions. Classical radiomics remains influential because its features are compact and often easier to audit than latent deep embeddings [23, 24, 46, 91, 130, 134, 184, 237].

The same design also creates fragility. A radiomic signature may depend on a specific scanner, reconstruction kernel, segmentation protocol, or cohort. Small sample sizes encourage overfitting, especially when thousands of features are screened before modeling. Robust radiomics therefore requires repeatability analysis, dimensionality control, external validation, transparent preprocessing, and careful endpoint definition. Nomograms that combine radiomics with clinical variables are common because imaging rarely explains prognosis alone [23, 24, 46, 91, 130, 134, 184, 237].

2.2 Deep Learning for Tumor Imaging

Deep learning changes the representation problem. Convolutional networks, transformers, self-supervised encoders, and foundation-style imaging models can learn hierarchical features from slices, volumes, patches, regions, or whole-slide images. These models can share representations across segmentation, detection, classification, grading, survival, and response tasks. They can also use weak labels, multiple instance learning, attention, and multimodal fusion to work with clinical data that are noisy or only partially annotated [24, 26, 46, 91, 121, 184, 211, 237].

The price of flexibility is reduced transparency and greater demand for data. Deep models can memorize institutional artifacts, acquisition protocols, or annotation conventions. Three-dimensional models are expensive and often trained on limited cohorts; two-dimensional models may miss volumetric context. Self-supervised and foundation models reduce label requirements, but they still need evaluation on clinically realistic endpoints and external cohorts. The field is therefore moving toward hybrid systems that combine radiomics features, deep embeddings, segmentation masks, clinical covariates, pathology, and genomics [24, 26, 46, 91, 121, 184, 211, 237].

2.3 Clinical Endpoints: Diagnosis, Prognosis, and Response

Diagnosis asks whether a lesion is present, malignant, high-grade, metastatic, or a specific subtype. Prognosis asks how long a patient is likely to survive, remain progression-free, relapse-free, or avoid recurrence. Response prediction asks whether a patient will benefit from a therapy or develop toxicity. These endpoints demand different supervision. A diagnostic classifier may use cross-sectional labels; a prognostic model must handle censoring and competing risks; a response model needs treatment-specific labels and timing. Endpoint-aware design is therefore essential [23, 24, 27, 46, 130, 134, 184, 237].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined radiomics, deep learning, tumor or cancer imaging, diagnosis, prognosis, survival, recurrence, treatment response, radiotherapy, chemotherapy, immunotherapy, neoadjuvant therapy, CT radiomics, MRI radiomics, PET radiomics, ultrasound, mammography, brain tumor, glioma, lung cancer, breast cancer, liver cancer, prostate cancer, colorectal cancer, head and neck cancer, radiogenomics, pathomics, histopathology, multimodal fusion, clinical data, self-supervised cancer imaging, external validation, robustness, interpretability, and calibration. Records were retained when title or abstract evidence explicitly connected tumor or cancer imaging with radiomics, deep learning, diagnosis, prognosis, survival, recurrence, or therapy-response modeling. Duplicates were merged by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *representation*: handcrafted radiomics features, deep embeddings, segmentation masks, pathology patches, genomic profiles, clinical variables, or multimodal patient vectors. The second is *imaging modality*: CT, MRI, PET, ultrasound, mammography, digital pathology, or hybrid modalities. The third is *tumor site*: brain, lung, breast, liver, prostate, colorectal, rectal, head and neck, pancreatic, renal, ovarian, and other cancers. The fourth is *endpoint*: diagnosis, grading, subtype classification, staging, survival, recurrence, progression, treatment response, toxicity, or therapy planning. The fifth is *model family*: linear and Cox models, random forests, support-vector machines, gradient boosting, CNNs, transformers, graph models, multiple-instance learning, and self-supervised encoders. The sixth is *validation design*: internal split, cross-validation, temporal validation, external cohort, multicenter study, prospective evaluation, or clinical workflow analysis. The seventh is *interpretability*: feature importance, attention, saliency, counterfactuals, biological pathway links, and radiologist review. The eighth is *deployment*: robustness, calibration, fairness, data governance, segmentation burden, and integration into oncology decision making.

3.3 Feature Engineering and Classical Radiomics

Classical radiomics is a modeling pipeline rather than a single algorithm. Its strength is modularity: segmentation, preprocessing, feature extraction, feature filtering, model fitting, calibration, and nomogram construction can each be audited. This is useful in small clinical cohorts where a fully end-to-end deep model may overfit. Feature families such as intensity histograms, gray-level co-occurrence matrices, run-length matrices, shape descriptors, Laplacian-of-Gaussian filters, and wavelets can capture tumor heterogeneity and margin structure. But they should be used with stability checks and feature-selection discipline [23, 24, 46, 91, 130, 134, 184, 237].

3.4 Deep Tumor Representations

Deep learning methods can learn spatial phenotypes that engineered features may miss. In diagnosis, networks detect or classify lesions from image patches, slices, or volumes. In prognosis, encoders generate representations that feed Cox, DeepSurv, discrete-time survival, ranking, or risk-stratification heads. In response prediction, models combine baseline and follow-up scans or learn biomarkers of sensitivity to therapy. Transformer and self-supervised methods are particularly important when annotations are scarce or heterogeneous across institutions [24, 26, 46, 91, 121, 184, 211, 237].

3.5 Diagnosis, Grading, and Classification

Diagnostic systems focus on lesion detection, benign-versus-malignant discrimination, tumor subtype classification, grading, staging support, and differential diagnosis. They often benefit from segmentation because lesion boundaries and peritumoral context contain discriminative information. Diagnostic evaluation should report sensitivity, specificity, AUROC, decision thresholds, reader comparison, and subgroup performance. A model that performs well on a curated test set may still fail if deployed on screening data, incidental findings, rare subtypes, or images from different scanners [91, 106, 121, 134, 184, 211, 216, 237].

3.6 Prognosis, Survival, and Recurrence

Prognostic imaging AI attempts to estimate risk before clinical outcomes are observed. The endpoint may be overall survival, disease-free survival, progression-free survival, recurrence, metastasis, local control, or treatment toxicity. These tasks require censored survival modeling, time horizons, calibration, and risk-group interpretation. Imaging features can capture tumor burden and heterogeneity, but prognosis also depends on molecular subtype, treatment, performance status, comorbidities, and clinical management. Strong studies therefore integrate imaging with clinical and biological covariates rather than treating images as a complete prognostic record [23, 24, 27, 46, 130, 134, 184, 237].

3.7 Treatment Response and Therapy Planning

Response prediction is one of the most clinically valuable but hardest tasks. Baseline scans may reveal phenotypes associated with sensitivity to chemotherapy, immunotherapy, radiotherapy, targeted therapy, or neoadjuvant treatment. Longitudinal scans can measure early response, pseudo-progression, or residual disease. Models must distinguish generic prognosis from treatment-specific benefit; otherwise a high-risk score may not indicate which therapy should change. Response studies should define therapy, timing, label criteria, and competing clinical decisions explicitly [23, 24, 27, 105, 106, 121, 134, 211].

3.8 Radiogenomics, Pathomics, and Multimodal Fusion

Tumor imaging is one view of disease. Radiogenomics connects imaging phenotypes to molecular alterations, expression programs, immune state, and pathway activity. Pathomics uses histopathology to capture cellular morphology and microenvironment. Multimodal models combine imaging, pathology, genomics, clinical variables, and sometimes text. These approaches are promising because diagnosis and prognosis are multifactorial, but they must handle missing modalities, small paired cohorts, leakage between patients, and the difference between correlation and actionable biology [23, 24, 46, 91, 106, 121, 130, 134].

3.9 Modalities, Tumor Sites, and Segmentation Pipelines

Different modalities expose different tumor biology. CT is widely available and important for lung, liver, colorectal, pancreatic, and head-and-neck cancers. MRI offers soft-tissue contrast for brain, prostate, rectal, breast, and liver tumors. PET captures metabolic activity. Ultrasound and mammography support screening and targeted diagnosis. Histopathology adds cellular resolution. Segmentation can be manual, semi-automatic, or learned; errors in segmentation propagate into both radiomics features and deep models, especially for small lesions and peritumoral regions [23, 46, 91, 130, 134, 184, 216, 237].

3.10 Validation, Robustness, and Clinical Translation

Clinical translation depends on evidence beyond aggregate accuracy. External validation tests whether a model survives new institutions, scanners, protocols, populations, and annotation styles. Multicenter studies can reveal brittleness hidden by random splits. Calibration matters because risk scores inform treatment and surveillance. Interpretability matters because clinicians need to know whether a model uses plausible tumor signal or confounded artifacts. Deployment also requires reporting preprocessing, segmentation requirements, missing-data handling, uncertainty, failure modes, and workflow integration [26, 91, 105, 121, 130, 184, 211, 237].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across radiomics feature extraction, deep tumor imaging, diagnosis and grading, survival and recurrence, treatment response, multimodal radiogenomics and pathomics, imaging modalities and tumor sites, and clinical validation.

Radiomics feature extraction and classical machine learning The radiomics feature extraction and classical machine learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [61, 67, 71, 130, 136, 184, 211, 223, 237]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [29, 37, 43, 77, 95, 103, 104, 123, 207]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [36, 44, 60, 88, 181, 192, 224, 252, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [38, 39, 93, 153, 188, 201, 243, 270, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 14, 45, 150, 167, 196, 221, 239, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 59, 75, 154, 171, 175, 199, 219, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 30, 35, 52, 113, 157, 247, 279, 286]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [28, 31, 34, 56, 98, 168, 170, 203, 208]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 107].

Deep learning for tumor imaging The deep learning for tumor imaging cluster contains work that develops methods, systems, datasets, or analyses central to this survey [91, 114, 156, 161, 169, 173, 185, 193, 246]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [4, 7, 10, 76, 133, 204, 238, 245, 265]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 22, 78, 100, 198, 220, 255, 272, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [90, 140, 143, 176, 180, 194, 267, 278, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [17, 101, 127, 147, 205, 266, 268, 271, 280]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [68, 69, 81, 108, 132, 141, 166, 233, 235]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 115, 159, 174, 177].

Tumor diagnosis, grading, and classification The tumor diagnosis, grading, and classification cluster contains work that develops methods, systems, datasets, or analyses central to this survey [5, 20, 54, 63, 112, 134, 216, 260, 296]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [70, 86, 124, 142, 144, 213, 225, 232, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [62, 82, 117, 182, 186, 227, 228, 258, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 85, 96, 102, 155, 158, 285, 287, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [84, 99, 189, 195, 209, 212, 229, 253, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 80, 111, 116, 163, 164, 254].

Prognosis, survival, and recurrence prediction The prognosis, survival, and recurrence prediction cluster contains work that develops methods, systems, datasets, or analyses central to this survey [8, 25–27, 46, 72, 79, 106, 214]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [18, 92, 126, 165, 210, 218, 240, 251, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 42, 118, 135, 172, 178, 179, 277, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21, 119, 217, 248, 256, 259, 261, 263, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [51, 97, 125, 152, 202, 215, 236, 276, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [48].

Treatment response and therapy planning The treatment response and therapy planning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [23, 50, 105, 200, 230, 249].

Multimodal radiogenomics and pathomics The multimodal radiogenomics and pathomics cluster contains work that develops methods, systems, datasets, or analyses central to this survey [24, 47, 121, 139, 190, 231, 242, 257, 282]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [1, 33, 49, 55, 73, 89, 110, 131, 234]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [87, 250, 298].

Modalities, tumor sites, and segmentation pipelines The modalities, tumor sites, and segmentation pipelines cluster contains work that develops methods, systems, datasets, or analyses central to this survey [9, 32, 53, 66, 74, 148, 160, 191, 290]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [58, 64, 65, 120, 146, 151, 183, 241, 244]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [41, 109, 128, 129, 162, 197, 206, 226, 262]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [16, 122, 137, 145, 149, 187, 222, 269, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [40, 94, 138, 264].

Validation, robustness, and clinical deployment The validation, robustness, and clinical deployment cluster contains work that develops methods, systems, datasets, or analyses central to this survey [83, 274].

4 Challenges and Prospects

4.1 Reproducibility Across Acquisition Pipelines

Radiomic features and deep embeddings can change with scanner vendor, reconstruction, slice thickness, contrast timing, motion, and preprocessing. Future studies need harmonization, test-retest analysis, acquisition metadata, and external cohorts that expose protocol variation.

4.2 Endpoint-Aware Survival and Response Modeling

Diagnosis, prognosis, and response are not interchangeable labels. Prognostic studies need censored survival methods and calibrated time horizons; response studies need treatment-specific designs that separate predictive biomarkers from generic risk markers.

4.3 Segmentation Burden and Error Propagation

Many pipelines assume accurate tumor contours, but segmentation is costly and variable. Robust systems should quantify how contour uncertainty affects features, embeddings, and clinical decisions, and should support weakly supervised or joint segmentation-prediction designs where appropriate.

4.4 Multimodal Missingness and Confounding

Radiogenomics and pathomics can improve biological grounding, but paired imaging, pathology, genomics, and outcomes are often incomplete. Multimodal models must handle missing views without hallucinating evidence and should guard against confounding by treatment, stage, institution, and patient selection.

4.5 Calibration, Interpretability, and Clinical Governance

Tumor imaging AI should produce calibrated probabilities or risks, not only rankings. Interpretability should connect model behavior to plausible tumor regions, radiomic features, histologic patterns, or molecular mechanisms. Governance requires versioned models, audit trails, subgroup monitoring, and clear policies for human oversight.

5 Conclusion

Radiomics and deep learning provide complementary routes from tumor images to quantitative oncology biomarkers. Radiomics offers compact, auditable descriptors; deep learning offers flexible representation learning; multimodal models connect imaging to pathology, genomics, clinical variables, and therapy context. The field's next step is not simply larger models, but stronger evidence: external validation, endpoint-aware survival and response modeling, calibrated risk estimates, robust segmentation, interpretable biomarkers, and clinical workflows that make uncertainty visible.

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